Multivariate Location Tests: Inferences about a Mean Vector

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Content adapted from:
Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed).

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Univariate Reminder: Student's One-Sample t Test

Let $\{x_1, \ldots, x_n\}$ denote a sample of iid observations sampled from a normal distribution with mean μ and variance σ^2, i.e., $x_i \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2)$.

Suppose σ^2 is unknown, and we want to test the hypotheses

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

where μ_0 is some known value specified by the null hypothesis.

We use Student's t test, where the t test statistic is given by

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})^2$.

Univariate Reminder: Student's t Test (continued)

Under H_0, the t test statistic follows a Student's t distribution with degrees of freedom $\nu = n - 1$.

- We reject H_0 if $|t|$ is large relative to what we would expect
- Same as rejecting H_0 if t^2 is larger than we expect

We can rewrite the (squared) t test statistic as

$$t^2 = n(\bar{x} - \mu_0)(s^2)^{-1}(\bar{x} - \mu_0)$$

which emphasizes the quadratic form of the t test statistic.

- t^2 gets larger as $|\bar{x} - \mu_0|$ gets larger (for fixed s^2 and n)
- t^2 gets larger as s^2 gets smaller (for fixed $\bar{x} - \mu_0$ and n)
- t^2 get larger as n gets larger (for fixed $\bar{x} - \mu_0$ and s^2)

Multivariate Extensions of Student's t Test

Now suppose that $\mathbf{x}_i \overset{\text{iid}}{\sim} \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

- $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$ is the i-th observation's $p \times 1$ vector
- $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)^\top$ is the $p \times 1$ mean vector
- $\boldsymbol{\Sigma} = \{\sigma_{jk}\}$ is the $p \times p$ covariance matrix

Suppose $\boldsymbol{\Sigma}$ is unknown, and we want to test the hypotheses

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \quad \text{versus} \quad H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

where $\boldsymbol{\mu}_0$ is some known vector specified by the null hypothesis.

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Hotelling's T^2 Test Statistic

Hotellings T^2 is multivariate extension of (squared) t test statistic

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$$

where

- $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$ is the sample mean vector
- $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is the sample covariance matrix
- $\frac{1}{n}\mathbf{S}$ is the sample covariance matrix of $\bar{\mathbf{x}}$

Letting $\mathbf{X} = \{x_{ij}\}$ denote the $n \times p$ data matrix, we could write

- $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i = n^{-1}\mathbf{X}^\top \mathbf{1}_n$
- $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = \frac{1}{n-1}\mathbf{X}_c^\top \mathbf{X}_c$
- $\mathbf{X}_c = \mathbf{C}\mathbf{X}$ with $\mathbf{C} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ denoting a centering matrix

Inferences using Hotelling's T^2

Under H_0, Hotelling's T^2 follows a scaled F distribution

$$T^2 \sim \frac{(n-1)p}{(n-p)} F_{p,n-p}$$

where $F_{p,n-p}$ denotes an F distribution with p numerator degrees of freedom and $n-p$ denominator degrees of freedom.

- This implies that $\alpha = P(T^2 > [p(n-1)/(n-p)]F_{p,n-p}(\alpha))$
- $F_{p,n-p}(\alpha)$ denotes upper (100α)th percentile of $F_{p,n-p}$ distribution

We reject the null hypothesis if T^2 is too large, i.e., if

$$T^2 > \frac{(n-1)p}{(n-p)} F_{p,n-p}(\alpha)$$

where α is the significance level of the test.

Comparing Student's t^2 and Hotelling's T^2

Student's t^2 and Hotelling's T^2 have a similar form

$$\begin{aligned} T^2_{p,n-1} &= \sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top [\mathbf{S}]^{-1} \sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \\ &= (\text{MVN vector})^\top \left(\frac{\text{Wishart matrix}}{df} \right)^{-1} (\text{MVN vector}) \end{aligned}$$

$$\begin{aligned} t^2_{n-1} &= \sqrt{n}(\bar{x} - \mu_0)[s^2]^{-1} \sqrt{n}(\bar{x} - \mu_0) \\ &= (\text{UVN variable}) \left(\frac{\text{scaled } \chi^2 \text{ variable}}{df} \right)^{-1} (\text{UVN variable}) \end{aligned}$$

where MVN (UVN) = multivariate (univariate) normal and $df = n - 1$.

Define Hotelling's T^2 Test Function

```
T.test <- function(X, mu = 0)\{
  X <- as.matrix(X)
  n <- nrow(X)
  p <- ncol(X)
  df2 <- n - p
  if(df2 < 1L) stop("Need nrow(X) > ncol(X).")
  if(length(mu) != p) mu <- rep(mu[1], p)
  xbar <- colMeans(X)
  S <- cov(X)
  T2 <- n * t(xbar - mu) %*% solve(S) %*% (xbar - mu)
  Fstat <- T2 / (p * (n-1) / df2)
  pval <- 1 - pf(Fstat, df1 = p, df2 = df2)
  data.frame(T2 = as.numeric(T2), Fstat = as.numeric(Fstat),
             df1 = p, df2 = df2, p.value = as.numeric(pval), row.names="")
\}
```

Hotelling's T^2 Test Example in R

```
# get data matrix
> data(mtcars)
> X <- mtcars[,c("mpg","disp","hp","wt")]
> xbar <- colMeans(X)

# try Hotelling T^2 function
> xbar
      mpg      disp        hp        wt
 20.09062 230.72188 146.68750   3.21725
> T.test(X)
      T2    Fstat df1 df2 p.value
 7608.14 1717.967   4  28       0
> T.test(X, mu = c(20, 200, 150, 3))
       T2    Fstat df1 df2    p.value
 10.78587 2.435519   4  28 0.07058328
> T.test(X, mu=xbar)
 T2 Fstat df1 df2 p.value
  0     0   4  28       1
```

Hotelling's T^2 Using `lm` Function in R

```
> y <- as.matrix(X)
> anova(lm(y ~ 1))
Analysis of Variance Table

            Df  Pillai approx F num Df den Df    Pr(>F)
(Intercept)  1 0.99594     1718      4     28 < 2.2e-16 ***
Residuals   31
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


> y <- as.matrix(X) - matrix(c(20,200,150,3),nrow(X),ncol(X),byrow=T)
> anova(lm(y ~ 1))
Analysis of Variance Table

            Df  Pillai approx F num Df den Df  Pr(>F)
(Intercept)  1 0.25812   2.4355      4     28 0.07058 .
Residuals   31
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


> y <- as.matrix(X) - matrix(xbar,nrow(X),ncol(X),byrow=T)
> anova(lm(y ~ 1))
Analysis of Variance Table

            Df     Pillai   approx F num Df den Df Pr(>F)
(Intercept)  1 1.8645e-31 1.3052e-30      4     28      1
Residuals   31
```

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Multivariate Normal MLE Reminder

Reminder: the log-likelihood function for n independent samples from a p-variate normal distribution has the form

$$LL(\boldsymbol{\mu}, \boldsymbol{\Sigma}|\mathbf{X}) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}\sum_{i=1}^{n}(\mathbf{x}_i - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu})$$

and the MLEs of the mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ are

$$\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i = \bar{\mathbf{x}}$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top$$

Multivariate Normal Maximized Likelihood Function

Plugging the MLEs of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ into the likelihood function gives

$$\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu},\boldsymbol{\Sigma}|\mathbf{X}) = \frac{\exp(-np/2)}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}|^{n/2}}$$

If we assume that $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ under H_0, then we have that

$$\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\Sigma}|\boldsymbol{\mu}_0,\mathbf{X}) = \frac{\exp(-np/2)}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}_0|^{n/2}}$$

where $\hat{\boldsymbol{\Sigma}}_0 = \frac{1}{n}\sum_{i=1}^n(\mathbf{x}_i - \boldsymbol{\mu}_0)(\mathbf{x}_i - \boldsymbol{\mu}_0)^\top$ is the MLE of $\boldsymbol{\Sigma}$ under H_0.

Likelihood Ratio Test Statistic (and Wilks' lambda)

Want to test $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$

The likelihood ratio test statistic is

$$\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\Sigma} | \boldsymbol{\mu}_0, \mathbf{X})}{\max_{\boldsymbol{\mu}, \boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{X})} = \left(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} \right)^{n/2}$$

and we reject H_0 if the observed value of Λ is too small.

The equivalent test statistic $\Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}$ is known as Wilks' lambda.

Relationship Between T^2 and Λ

There is a simple relationship between T^2 and Λ

$$\Lambda^{2/n} = \left(1 + \frac{T^2}{n-1}\right)^{-1}$$

which derives from the definition of the matrix determinant.[1]

This implies that we can just use the T^2 distribution for Λ inference.

- Reject H_0 for small $\Lambda^{2/n} \iff$ large T^2

[1] For a proof, see p 218 of Johnson & Wichern (2007).

General Likelihood Ratio Tests

Let $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ denote a $p \times 1$ vector of parameters, which takes values in the parameter set $\boldsymbol{\Theta}$, and let $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta}_0$ where $\boldsymbol{\Theta}_0 \subset \boldsymbol{\Theta}$.

A likelihood ratio test rejects $H_0 : \boldsymbol{\theta} \in \boldsymbol{\Theta}_0$ in favor of $H_1 : \boldsymbol{\theta} \notin \boldsymbol{\Theta}_0$ if

$$\Lambda = \frac{\max_{\boldsymbol{\theta} \in \boldsymbol{\Theta}_0} L(\boldsymbol{\theta})}{\max_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} L(\boldsymbol{\theta})} < c_\alpha$$

where c_α is some constant and $L(\cdot)$ is the likelihood function.

For a large sample size n, we have that

$$-2\log(\Lambda) \approx \chi^2_{\nu - \nu_0}$$

where ν and ν_0 are the dimensions of $\boldsymbol{\Theta}$ and $\boldsymbol{\Theta}_0$.

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Extending Confidence Intervals to Regions

A $100(1-\alpha)\%$ confidence interval (CI) for $\theta \in \Theta$ is defined such that

$$P[L_\alpha(\mathbf{x}) \leq \theta \leq U_\alpha(\mathbf{x})] = 1 - \alpha$$

where the interval $[L_\alpha(\mathbf{x}), U_\alpha(\mathbf{x})] \subset \Theta$ is a function of the data vector $\mathbf{x}$ and the significance level α.

A confidence region is a multivariate extension of a confidence interval.

A $100(1-\alpha)\%$ confidence region (CR) for $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ is defined such that

$$P[\boldsymbol{\theta} \in R_\alpha(\mathbf{X})] = 1 - \alpha$$

where the region $R_\alpha(\mathbf{X}) \subset \boldsymbol{\Theta}$ is a function of the data matrix $\mathbf{X}$ and the significance level α.

Confidence Regions for Normal Mean Vector

Before we collect n samples from a p-variate normal distribution

$$P[T^2 \leq \nu_{n,p} F_{p,n-p}(\alpha)] = 1 - \alpha$$

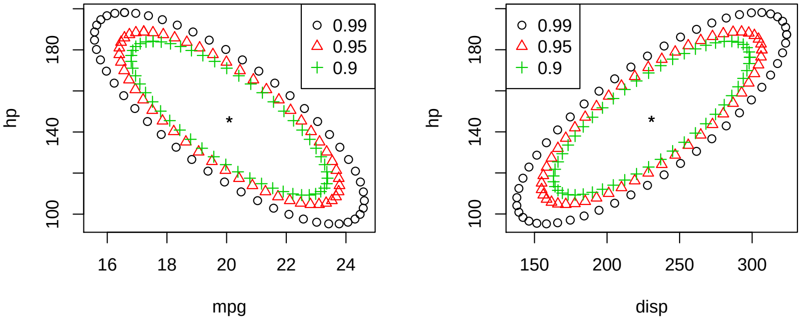
where $T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})$ and $\nu_{n,p} = p(n-1)/(n-p)$.

The $100(1-\alpha)\%$ confidence region (CR) for a mean vector from a p-variate normal distribution is ellipsoid formed by all $\boldsymbol{\mu} \in \mathbb{R}^p$ such that

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \leq \nu_{n,p} F_{p,n-p}(\alpha)$$

where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$ and $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.

Forming Confidence Regions in R




```
n <- nrow(X)
p <- ncol(X)
xbar <- colMeans(X)
S <- cov(X)
library(car)
tconst <- sqrt((p/n)*((n-1)/(n-p)) * qf(0.99,p,n-p))   # for 99% confidence region
id <- c(1,3)
plot(ellipse(center = xbar[id], shape = S[id,id], radius = tconst, draw = F), xlab = "mpg", ylab = "hp")
```

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Linear Combinations of Normal Variables

Suppose $\boldsymbol{X} = (X_1, \ldots, X_p)^\top \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{a} = (a_1, \ldots, a_p)^\top \in \mathbb{R}^p$ denote some linear transformation vector.

The random variable $Z = \sum_{j=1}^p a_j X_j = \mathbf{a}^\top \boldsymbol{X}$ has the properties

$$\mu_Z = E(Z) = \sum_{j=1}^p a_j E(X_j) = \mathbf{a}^\top \boldsymbol{\mu}$$
$$\sigma_Z^2 = \mathrm{Var}(Z) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$$

and because Z is a linear transformation of normal variables, we know

$$Z \sim \mathrm{N}(\mu_Z, \sigma_Z^2)$$

i.e., Z follows a univariate normal with mean μ_Z and variance σ_Z^2.

Linear Combinations of Multivariate Sample Means

Suppose that $\mathbf{x}_i \stackrel{\text{iid}}{\sim} \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $z_i = \mathbf{a}^\top \mathbf{x}_i$ for $i \in \{1, \ldots, n\}$.

The sample mean and variance of the z_i terms are

$$\bar{z} = \mathbf{a}^\top \bar{\mathbf{x}}$$
$$s_z^2 = \mathbf{a}^\top \mathbf{S} \mathbf{a}$$

where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and $\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.

Because $\bar{z}$ is a linear transformation of normal variables, we know

$$\bar{z} \sim \mathrm{N}(\mu_Z, \sigma_Z^2/n)$$

i.e., $\bar{z}$ follows a univariate normal with mean μ_Z and variance σ_Z^2/n.

Confidence Intervals for Single Linear Combination

Fixing $\mathbf{a}$ and assuming σ_Z^2 is unknown, the t test statistic is

$$t = \frac{\bar{z} - \mu_Z}{s_z/\sqrt{n}} = \frac{\sqrt{n}(\mathbf{a}^\top \bar{\mathbf{x}} - \mathbf{a}^\top \boldsymbol{\mu})}{\sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}}}$$

and the corresponding confidence interval has the form

$$\bar{z} - \frac{s_z}{\sqrt{n}} t_{n-1}(\alpha/2) \leq \mu_Z \leq \bar{z} + \frac{s_z}{\sqrt{n}} t_{n-1}(\alpha/2)$$

$$\mathbf{a}^\top \bar{\mathbf{x}} - \frac{\sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}}}{\sqrt{n}} t_{n-1}(\alpha/2) \leq \mu_Z \leq \mathbf{a}^\top \bar{\mathbf{x}} + \frac{\sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}}}{\sqrt{n}} t_{n-1}(\alpha/2)$$

where $t_{n-1}(\alpha/2)$ is the critical value that cuts off the upper $(100\alpha/2)\%$ tail of the t_{n-1} distribution.

Limitation of CIs for Single Linear Combination

The confidence interval has level α separately for each CI we form.

- Each interval separately satisfies $P[\mu_j \in R_j(\mathbf{X})] = 1 - \alpha$

For multiple CIs, the familywise significance level will exceed α.

- Intervals dot not satisfy $P[\mu_1 \in R_1(\mathbf{X}) \cap \cdots \cap \mu_p \in R_p(\mathbf{X})] = 1 - \alpha$

If we want to form a CI for each μ_j term, we need to consider some approach that will control the familywise error rate.

Defining a Simultaneous Confidence Interval

For a fixed $\mathbf{a}$, the t test statistic CI is the set of $\mathbf{a}^\top \boldsymbol{\mu}$ values such that

$$t^2 = \frac{(\bar{z} - \mu_Z)^2}{s_z^2/n} = \frac{n(\mathbf{a}^\top \bar{\mathbf{x}} - \mathbf{a}^\top \boldsymbol{\mu})^2}{\mathbf{a}^\top \mathbf{S} \mathbf{a}} \leq |t_{n-1}(\alpha/2)|^2$$

For a simultaneous CI, we want the above to hold for all choices of $\mathbf{a}$.

Start by considering the maximum possible t^2 that we could see

$$\max_{\mathbf{a}} t^2 = \max_{\mathbf{a}} \frac{n[\mathbf{a}^\top (\bar{\mathbf{x}} - \boldsymbol{\mu})]^2}{\mathbf{a}^\top \mathbf{S} \mathbf{a}} = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}) = T^2$$

which occurs when $\mathbf{a} \propto \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})$.

Simultaneous Confidence Intervals Based on T^2

The fact that $\max_{\mathbf{a}} t^2 = T^2$ leads to the following simultaneous CI

$$\bar{z} - \frac{s_z}{\sqrt{n}}\sqrt{\nu_{n,p}F_{p,n-p}(\alpha)} \leq \mu_Z \leq \bar{z} + \frac{s_z}{\sqrt{n}}\sqrt{\nu_{n,p}F_{p,n-p}(\alpha)}$$

$$\mathbf{a}^\top\bar{\mathbf{x}} - \frac{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}}}{\sqrt{n}}\sqrt{\nu_{n,p}F_{p,n-p}(\alpha)} \leq \mu_Z \leq \mathbf{a}^\top\bar{\mathbf{x}} + \frac{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}}}{\sqrt{n}}\sqrt{\nu_{n,p}F_{p,n-p}(\alpha)}$$

which uses the fact that $T^2 \sim \nu_{n,p}F_{p,n-p}$ where $\nu_{n,p} = p(n-1)/(n-p)$.

Simultaneously for all linear combination vectors $\mathbf{a} \in \mathbb{R}^p$, the above interval will contain $\mathbf{a}^\top\boldsymbol{\mu}$ with probability $1-\alpha$.

R Function for Simultaneous T^2 Confidence Intervals

```
T.ci <- function(mu, Sigma, n, avec = rep(1, length(mu)), level = 0.95)\{
  p <- length(mu)
  if(nrow(Sigma) != p) stop("Need length(mu) == nrow(Sigma)")
  if(ncol(Sigma) != p) stop("Need length(mu) == ncol(Sigma)")
  if(length(avec) != p) stop("Need length(mu) == length(avec)")
  if((level <= 0) | (level >= 1)) stop("Need 0 < level < 1")
  cval <- qf(level, p, n-p) * p * (n-1) / (n-p)
  zhat <- crossprod(avec, mu)
  zvar <- crossprod(avec, Sigma %*% avec) / n
  const <- sqrt(cval * zvar)
  c(lower = zhat - const, upper = zhat + const)
\}
```

Example of Simultaneous T^2 Confidence Intervals

```
> X <- mtcars[,c("mpg","disp","hp","wt")]
> n <- nrow(X)
> p <- ncol(X)
> xbar <- colMeans(X)
> S <- cov(X)
> xbar
      mpg      disp        hp        wt
 20.09062 230.72188 146.68750   3.21725
> T.ci(mu = xbar, Sigma = S, n = n, avec = c(1,0,0,0))
   lower    upper
16.39689 23.78436
> T.ci(mu = xbar, Sigma = S, n = n, avec = c(0,1,0,0))
   lower    upper
154.7637 306.6801
> T.ci(mu = xbar, Sigma = S, n = n, avec = c(0,0,1,0))
   lower    upper
104.6674 188.7076
> T.ci(mu = xbar, Sigma = S, n = n, avec = c(0,0,0,1))
   lower    upper
2.617584 3.816916
```

Compare Simultaneous T^2 CI to Classic t CI

```
TCI <- tCI <- NULL
for(k in 1:4)\{
  avec <- rep(0, 4)
  avec[k] <- 1
  TCI <- c(TCI, T.ci(xbar, S, n, avec))
  tCI <- c(tCI,
           xbar[k] - sqrt(S[k,k]/n) * qt(0.975, df=n-1),
           xbar[k] + sqrt(S[k,k]/n) * qt(0.975, df=n-1))
\}
rtab <- rbind(TCI, tCI)

> round(rtab, 2)
    mpg.lower mpg.upper disp.lower disp.upper
TCI     16.40     23.78     154.76     306.68
tCI     17.92     22.26     186.04     275.41
    hp.lower hp.upper wt.lower wt.upper
TCI   104.67   188.71     2.62     3.82
tCI   121.97   171.41     2.86     3.57
```

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More Precise Simultaneous Confidence Intervals

If p and/or the number of linear combinations $\mathbf{a}_1, \ldots, \mathbf{a}_q$ is small, we may be able to form better (i.e., narrower) simultaneous CIs.

Let C_k denote some confidence statement, and note that

$$\begin{aligned} P[\text{all } C_k \text{ true}] &= 1 - P[\text{at least one } C_k \text{ false}] \\ &\geq 1 - \sum_{k=1}^{q} P(C_k \text{ false}) \\ &= 1 - \sum_{k=1}^{q} \alpha_k \end{aligned}$$

where α_k is the significance level for the k-th test.

Simultaneous CIs via Bonferroni's Method

Result on the previous slide is a special case of Bonferroni's inequality.

To control familywise error rate, we just need to adjust the error rates of the individual tests, i.e., the α_k terms.

If no prior knowledge is available,[2] we simply set $\alpha_k = \alpha/q$ to control the familywise error rate at α when conducting q significance tests.

- $100(1-\alpha)\%$ CI for q tests: $\quad \bar{z} \pm (s_z/\sqrt{n})t_{n-1}(\alpha_k/2)$ with $\alpha_k = \alpha/q$

[2] If we have prior knowledge about the importance of the individual tests, we could adjust each α_k individually with the constraint that $\sum_{k=1}^{q} \alpha_k = \alpha$.

Simultaneous CIs via Bonferroni's Method in R

```
TCI <- tCI <- bon <- NULL
alpha <- 1 - 0.05/(2*4)
for(k in 1:4)\{
  avec <- rep(0, 4)
  avec[k] <- 1
  TCI <- c(TCI, T.ci(xbar, S, n, avec))
  tCI <- c(tCI,
           xbar[k] - sqrt(S[k,k]/n) * qt(0.975, df=n-1),
           xbar[k] + sqrt(S[k,k]/n) * qt(0.975, df=n-1))
  bon <- c(bon,
           xbar[k] - sqrt(S[k,k]/n) * qt(alpha, df=n-1),
           xbar[k] + sqrt(S[k,k]/n) * qt(alpha, df=n-1))
\}
rtab <- rbind(TCI, tCI, bon)
> round(rtab, 2)
    mpg.lower mpg.upper disp.lower disp.upper
TCI     16.40     23.78     154.76     306.68
tCI     17.92     22.26     186.04     275.41
bon     17.27     22.92     172.62     288.82
    hp.lower hp.upper wt.lower wt.upper
TCI   104.67   188.71     2.62     3.82
tCI   121.97   171.41     2.86     3.57
bon   114.55   178.83     2.76     3.68
```

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Asymptotic Inference for Multivariate Means

Suppose that $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$ is a random sample from some distribution with finite mean $\boldsymbol{\mu}$ and finite covariance matrix $\boldsymbol{\Sigma}$.

As the sample size gets large, i.e, as $n \to \infty$, we have that

$$\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \approx \mathrm{N}(\mathbf{0}, \boldsymbol{\Sigma})$$
$$n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \approx \chi^2_p$$

where $\approx$ denotes "is approximately distributed as".

Result holds for non-normal data too, as long as $n - p$ is large!

Update T.test Function: Add asymp Option

```
T.test <- function(X, mu = 0, asymp = FALSE)\{
  X <- as.matrix(X)
  n <- nrow(X)
  p <- ncol(X)
  df2 <- n - p
  if(df2 < 1L) stop("Need nrow(X) > ncol(X).")
  if(length(mu) != p) mu <- rep(mu[1], p)
  xbar <- colMeans(X)
  S <- cov(X)
  T2 <- n * t(xbar - mu) %*% solve(S) %*% (xbar - mu)
  Fstat <- T2 / (p * (n-1) / df2)
  if(asymp)\{
    pval <- 1 - pchisq(T2, df = p)
  \} else \{
    pval <- 1 - pf(Fstat, df1 = p, df2 = df2)
  \}
  data.frame(T2 = as.numeric(T2), Fstat = as.numeric(Fstat),
             df1 = p, df2 = df2, p.value = as.numeric(pval),
             asymp = asymp, row.names = "")
\}
```

Compare Finite Sample and Large Sample p-values

```
# compare finite sample and large sample p-values (n=10)
> set.seed(1)
> Xtemp <- matrix(rnorm(10*4), 10, 4)
> T.test(Xtemp)
       T2     Fstat df1 df2   p.value asymp
 1.963739 0.3272899   4   6 0.8503944 FALSE
> T.test(Xtemp, asymp = TRUE)
       T2     Fstat df1 df2   p.value asymp
 1.963739 0.3272899   4   6 0.7424283  TRUE

# compare finite sample and large sample p-values (n=50)
> set.seed(1)
> Xtemp <- matrix(rnorm(50*4), 50, 4)
> T.test(Xtemp)
       T2    Fstat df1 df2   p.value asymp
 4.348226 1.020502   4  46 0.4067571 FALSE
> T.test(Xtemp, asymp = TRUE)
       T2    Fstat df1 df2   p.value asymp
 4.348226 1.020502   4  46 0.3609251  TRUE

# compare finite sample and large sample p-values (n=100)
> set.seed(1)
> Xtemp <- matrix(rnorm(100*4), 100, 4)
> T.test(Xtemp)
       T2   Fstat df1 df2   p.value asymp
 1.972616 0.47821   4  96 0.7516411 FALSE
> T.test(Xtemp, asymp = TRUE)
       T2   Fstat df1 df2   p.value asymp
 1.972616 0.47821   4  96 0.7407957  TRUE
```

Forming Large Sample Confidence Intervals

For large n, we have that $T^2 \approx \chi^2_p$, which implies

$$P[T^2 \leq \chi^2_p(\alpha)] \approx 1 - \alpha$$

where $\chi^2_p(\alpha)$ is the upper (100α)th percentile of the χ^2_p distribution.

The implied large sample CI has the form

$$\bar{z} - \frac{s_z}{\sqrt{n}}\sqrt{\chi^2_p(\alpha)} \leq \mu_Z \leq \bar{z} + \frac{s_z}{\sqrt{n}}\sqrt{\chi^2_p(\alpha)}$$

$$\mathbf{a}^\top \bar{\mathbf{x}} - \frac{\sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}}}{\sqrt{n}}\sqrt{\chi^2_p(\alpha)} \leq \mu_Z \leq \mathbf{a}^\top \bar{\mathbf{x}} + \frac{\sqrt{\mathbf{a}^\top \mathbf{S} \mathbf{a}}}{\sqrt{n}}\sqrt{\chi^2_p(\alpha)}$$

Forming Large Sample Confidence Intervals in R

```
chi <- NULL
for(k in 1:4)\{
  chi <- c(chi,
           xbar[k] - sqrt(S[k,k]/n) * sqrt(qchisq(0.95, df=p)),
           xbar[k] + sqrt(S[k,k]/n) * sqrt(qchisq(0.95, df=p)))
\}

> round(rtab, 2)
    mpg.lower mpg.upper disp.lower disp.upper
TCI     16.40     23.78     154.76     306.68
tCI     17.92     22.26     186.04     275.41
bon     17.27     22.92     172.62     288.82
chi     16.81     23.37     163.24     298.21
    hp.lower hp.upper wt.lower wt.upper
TCI   104.67   188.71     2.62     3.82
tCI   121.97   171.41     2.86     3.57
bon   114.55   178.83     2.76     3.68
chi   109.35   184.02     2.68     3.75
```

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Prediction Regions for Future Observations

Suppose $\mathbf{x}_i \stackrel{\text{iid}}{\sim} \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\bar{\mathbf{x}}$ and $\mathbf{S}$ have been calculated from a sample of n independent observations.

If $\mathbf{x}_*$ is some new observation sampled from $\mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

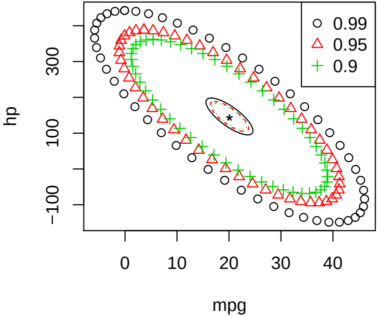
$$T_*^2 = \frac{n}{n+1}(\mathbf{x}_* - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_* - \bar{\mathbf{x}}) \sim \frac{(n-1)p}{n-p} F_{p,n-p}$$

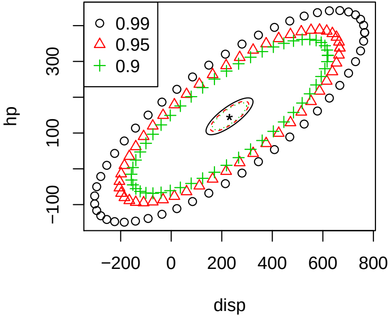
given that $\mathrm{Cov}(\mathbf{x}_* - \bar{\mathbf{x}}) = \mathrm{Cov}(\mathbf{x}_*) + \mathrm{Cov}(\bar{\mathbf{x}}) = [(n+1)/n]\boldsymbol{\Sigma}$.

The $100(1-\alpha)\%$ prediction ellipsoid is given by all $\mathbf{x}_*$ that satisfy

$$(\mathbf{x}_* - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_* - \bar{\mathbf{x}}) \leq \frac{(n^2-1)p}{n(n-p)} F_{p,n-p}(\alpha)$$

Forming Prediction Regions in R






```
n <- nrow(X)
p <- ncol(X)
xbar <- colMeans(X)
S <- cov(X)
library(car)
pconst <- sqrt((p/n)*((n^2-1)/(n-p)) * qf(0.99,p,n-p))
id <- c(1,3)
plot(ellipse(center=xbar[id], shape=S[id,id], radius=pconst, draw=F), xlab="mpg", ylab="hp")
```